

Does Satellite Data Match Air Quality on the Ground?

MODIS AOD as a Proxy for Ground-Level PM_{2.5} in Hyderabad, India

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Introduction

On an early morning from my balcony in Hyderabad, the buildings one kilometer away were still visible but looked faded. The observation raised a question: does the satellite data NASA publishes about aerosols over Hyderabad reflect what people are breathing?

Hyderabad's PM_{2.5} levels are consistently 5 to 6 times above the WHO annual guideline of 5 $\mu\text{g}/\text{m}^3$ (WHO, 2021). Satellite AOD from MODIS is used to fill gaps in India's ground monitoring network, which covers too little of the country relative to its population (Sathe et al., 2019). But AOD and PM_{2.5} measure different things. AOD captures the total aerosol load across the whole atmospheric column; PM_{2.5} measures only what is near the surface. Studies across Indian cities find weak correlations between the two without meteorological correction, with boundary layer dynamics as the main cause (Gupta et al., 2009; Sathe et al., 2019; Scaria et al., 2023).

This report tests whether MODIS Terra and Aqua AOD are reliable proxies for surface PM_{2.5} in Hyderabad during March and April 2026, and whether differences in their overpass times capture a pattern in how pollution is distributed through the atmosphere at different times of day.

Hypothesis: MODIS AOD and ground PM_{2.5} will show weak agreement in Hyderabad, because satellites measure the full atmospheric column while ground sensors measure only near-surface concentrations within the boundary layer.

Three predictions follow.

1. Neither Terra nor Aqua AOD will correlate significantly with daily PM_{2.5}.
2. Terra AOD will tend to be higher than Aqua AOD, because its earlier 10:30 IST overpass captures residual morning inversion conditions before afternoon mixing disperses near-surface aerosols.
3. Terra and Aqua will correlate strongly with each other, since both measure the same regional atmospheric signal.

Methods

Three quantitative continuous variables were collected from March 13 to April 12, 2026.

Ground PM_{2.5} ($\mu\text{g}/\text{m}^3$) came from Telangana State Pollution Control Board's Kokapet monitoring station, approximately 4 km from the balcony observation site (CPCB, 2026). Kokapet was chosen as the nearest station because it is not adjacent to a highway or a large green area, both of which would push results away from typical urban pollution levels. Daily 24-hour averages were used.

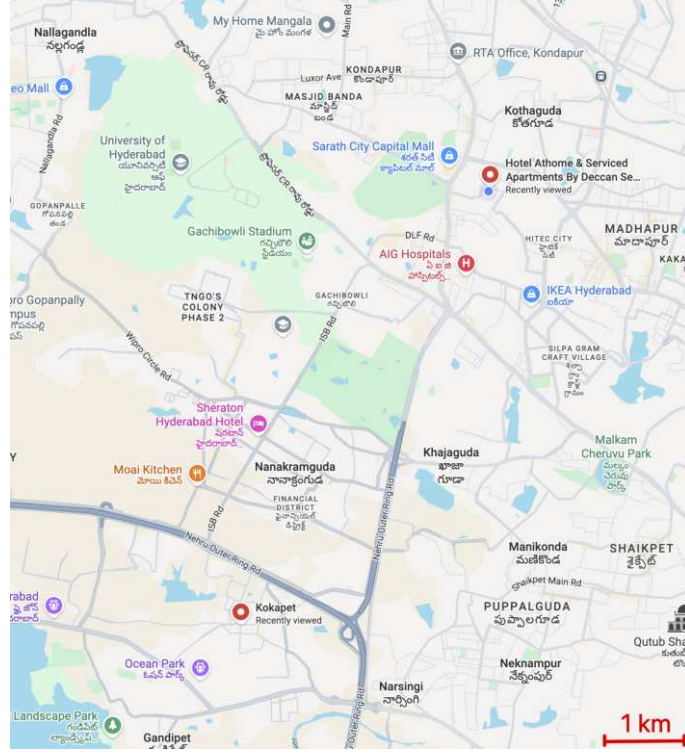


Figure 1: **Observation site and ground monitoring station locations, Hyderabad.** Both sites are marked with red-and-white-alternating circles on the map. They are approximately 6 km apart; the scale bar in the bottom right shows 1 km for reference.

Satellite AOD (unitless, 550 nm) was downloaded from NASA Giovanni (Giovanni, 2026) for both MODIS Terra (10:30 IST overpass, MOD08_D3; MODIS Terra, 2017) and MODIS Aqua (13:30 IST overpass, MYD08_D3; MODIS Aqua, 2017), using the Combined Dark Target and Deep Blue product, averaged over a 0.3° box centered on Kondapur.

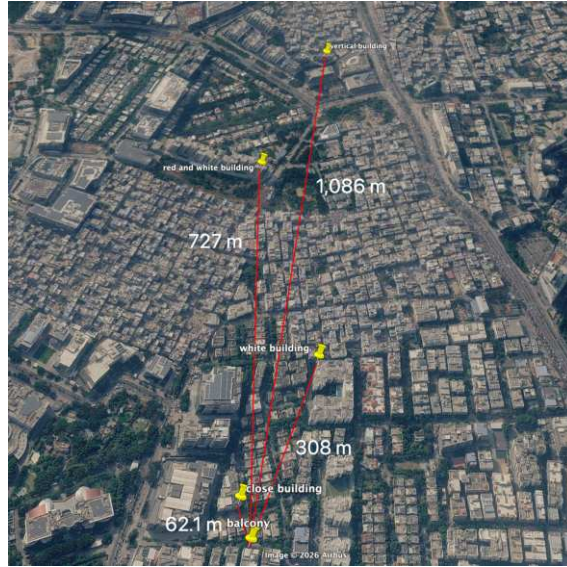
For the visual haze index, landmark-based visibility estimation was initially planned. Sight lines from the balcony were identified, and distances to visible features were measured using Google Earth Pro.



(a) Unlabeled balcony view, April 8.



(b) Labeled landmarks with distances, April 4.



(c) Bird's-eye view from Google Earth Pro.

Figure 2: Balcony observation site and landmark distances used in the initial visibility approach. (a) Balcony view with no labels, taken on April 8 (clearest day by contrast analysis). (b) Same view with labeled landmarks and measured distances. Close building: 62.1 m, white building: 308 m, red and white building: 727 m, vertical building: 1,086 m. Figures (a) and (b) are from different days, yet all landmarks remain clearly visible in both, illustrating why landmark-based estimation was not used as a proxy for visual haze index. (c) Bird's-eye view from Google Earth Pro with the same landmarks and distances marked.

Since all landmarks stayed visible on every observation day, this approach was excluded. A computational method was adopted instead: a Python script aligned each photo to a reference frame using SIFT keypoint matching (Lowe, 2004) and homography warping

in OpenCV (OpenCV, 2023), then computed the standard deviation of pixel brightness values within a fixed-sized patch at a given distance from the image center. This standard deviation, the visual haze index, measures how much contrast remains in distant features; haze reduces it by scattering light uniformly across the scene.



Figure 3: **Example output from the image alignment pipeline.** Left: reference photograph. Right: aligned target photograph. The green rectangle shows the distant region used for contrast measurement. The blue rectangle shows the near-field anchor used to verify alignment quality.

Photos were collected across 6 days (Table 1). Spearman rank correlations were used, given the small sample size.

Table 1: **Photo collection dates and times (IST).**

Date	Time (IST)
April 4	09:31, 18:25
April 5	10:12, 11:51, 16:56
April 6	14:23
April 8	11:07, 12:54
April 9	07:49, 11:50
April 10	07:54, 10:20

Results of Data Collection

PM_{2.5} ranged from 44 to 69 $\mu\text{g}/\text{m}^3$ and exceeded the WHO 24-hour guideline on all 24 days with data. AOD was consistently moderate across the 26-day satellite record. The visual haze index showed greater day-to-day variability than either satellite variable (Table 2).

Table 2: **Descriptive statistics for all three air quality variables.** Image contrast standard deviation is derived from 12 individual balcony photographs across 6 observation days (April 4–10, 2026). PM_{2.5} reflects daily 24-hour averages from the Kokapet TSPCB station (March 13 – April 12, 2026). Terra and Aqua AOD reflect daily area-averaged Level 3 MODIS retrievals over Kondapur for the same 31-day period, with 26 valid retrieval days out of 31 total.

Variable	n	Mean	SD	Min	Max
Image contrast std	12	53.38	3.83	45.04	61.97
PM _{2.5} daily mean ($\mu\text{g}/\text{m}^3$)	24	60.92	5.15	44.00	69.00
Terra AOD at 550 nm	26	0.152	0.010	0.110	0.166
Aqua AOD at 550 nm	26	0.146	0.006	0.135	0.157

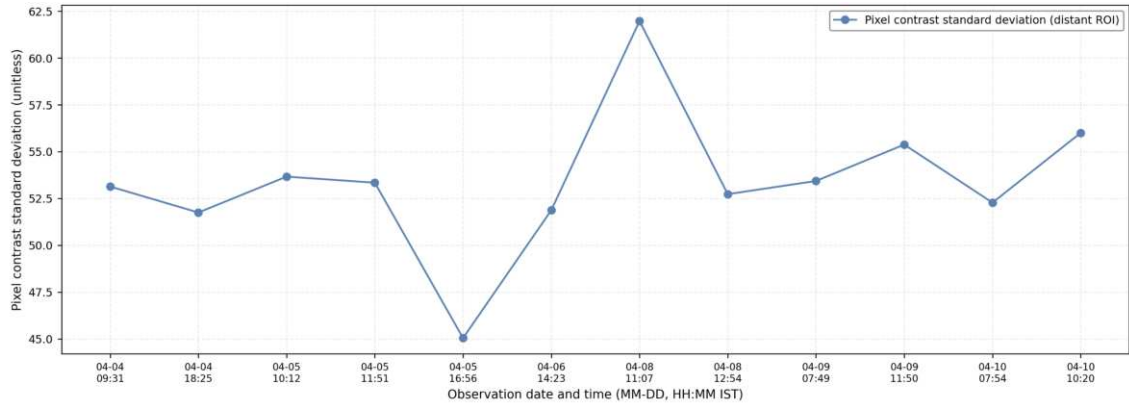


Figure 4: **Visual haze index over the observation period, Kondapur, April 2026.** Image contrast standard deviation from balcony photographs, April 4 to April 10, 2026 (IST). Each point is one photo. Higher values indicate clearer air. The lowest reading was April 5 at 16:56 IST (45.0); the highest was April 8 at 11:07 IST (62.0).



Figure 5: **Clearest and haziest balcony observations during the study period.** Clearest observation (left, April 8 11:07 IST, contrast std 62.0) and haziest observation (right, April 5 16:56 IST, contrast std 45.0). The visual difference is subtle, consistent with visibility remaining high throughout the study period.

Results

Terra AOD was higher than Aqua on most days (means 0.152 vs. 0.146), consistent with Prediction 2. The time series is shown in Figure 6. A direct comparison of AOD and PM2.5 magnitudes is not meaningful, as the variables have different units and physical meanings.

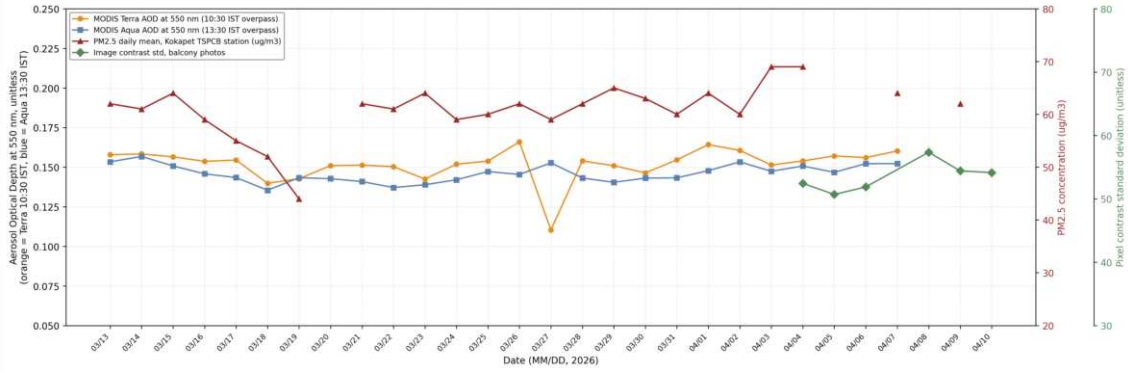


Figure 6: **Time series of all three air quality variables, Hyderabad, March to April 2026.** Daily Terra AOD (orange), Aqua AOD (blue), PM2.5 (red), and image contrast std (green) over the study period. AOD and PM2.5 cannot be directly compared in magnitude across axes; differences in scale reflect differences in what each variable measures.

Neither satellite AOD correlated significantly with PM2.5 (Figure 7). Terra and Aqua were strongly correlated. All three predictions held.

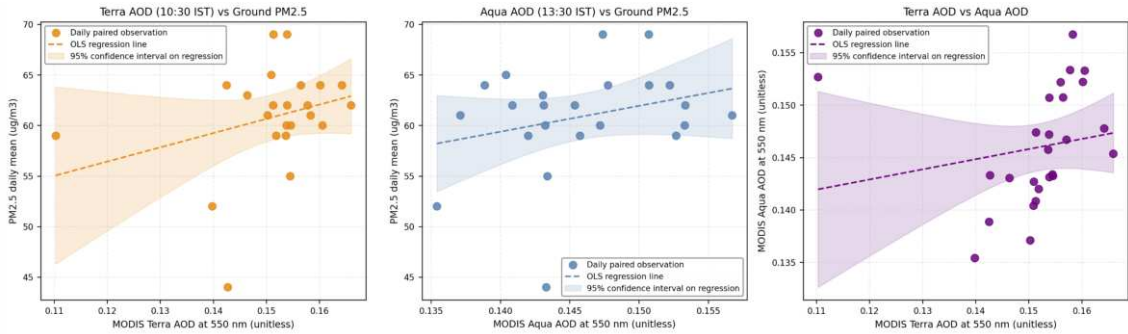


Figure 7: **Spearman rank correlations among air quality variables, Hyderabad, March to April 2026.** Left: Terra AOD vs. PM2.5 ($\rho = 0.233$, $p = 0.285$, $n = 23$). Centre: Aqua AOD vs. PM2.5 ($\rho = 0.128$, $p = 0.559$, $n = 23$). Right: Terra AOD vs. Aqua AOD ($\rho = 0.636$, $p = 0.001$, $n = 26$). Neither satellite overpass showed a significant relationship with ground PM2.5, while the two satellite products correlated significantly with each other.

The haze index had too few overlapping days with PM2.5 data ($n = 2$) for a formal test. The relationship was negative as expected (lower contrast on higher-PM2.5 days), but no statistical inference is possible at this sample size.

Discussion

All three predictions were confirmed. The weak AOD–PM2.5 correlations match findings from studies across Indian cities (Sathe et al., 2019; Scaria et al., 2023), and the Terra–Aqua correlation confirms that both instruments track the same regional aerosol signal.

AOD and PM2.5 would be expected to agree more on days when aerosols are well mixed throughout the boundary layer, since the vertical distribution would be roughly uniform

and surface concentrations would reflect the column total (Gupta et al., 2009). This is more likely in the afternoon, which is part of why the Aqua overpass is theoretically better matched to the daily average PM_{2.5}.

The disagreement is most likely due to elevated dust layers above the boundary layer, which inflate AOD without contributing to surface PM_{2.5}. The 1° spatial resolution of the Level 3 product also averages over a large area, including non-urban land, diluting the local signal.

Regardless of what the satellite shows, ground PM_{2.5} was above the WHO limit every single day. The most actionable local response is stricter enforcement of vehicle emission rules and dust suppression at construction sites, since transboundary dust transport is much harder to control.

For future work, three things would help most: a longer observation period (at least 60–90 days) to obtain more paired observations, boundary-layer height data to correct AOD for mixing depth, and a second camera site in a different district to give the haze index spatial meaning.

Code and Data

All code, raw data, and output files for this project are available at <https://drive.google.com/drive/folders/1q32rCsOxwICYGRmDa0jKb5cGDE7-ipCV?usp=sharing>.

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